INSURANCE COMPLAINT PATTERN & RESOLUTION ANALYSIS USING PYTHON (USA 2021-2026)

Exploratory and Statistical Investigation of Complaint Patterns, Resolution Speed, and Financial Recovery in U.S. Insurance Complaints (2021–2026)

Project Documentation

Python | Pandas | NumPy | Matplotlib | Seaborn | Google Colab

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Dataset	Insurance Company Complaints, Resolutions, Status, and Recoveries (data.gov)
Sample analysed	500 complaint records, 12 original columns
Analysis environment	Google Colab (Python 3)

1. Project Overview

This project analyzes an insurance company complaints dataset to understand complaint patterns, resolution trends and recovery behavior. Using Python, the analysis examines complaint volumes by company, coverage type and reason, how long complaints take to resolve, and how much money was recovered for consumers.

The project applies data inspection, data cleaning and transformation, feature engineering, statistical analysis and data visualization to convert raw complaint records into meaningful findings and business recommendations. The completed work is exploratory and statistical; it does not include a machine-learning prediction model.

The project follows a complete data analytics pipeline: **Data Loading** → **Initial EDA** → **Data Cleaning** → **Transformation** → **Feature Engineering** → **Visualization** → **Key Insights** → **Conclusion.**

2. Project Objective

To analyze insurance complaint data using Python by exploring complaint companies, coverage types, complaint reasons, resolution times and recovery amounts. The project aims to identify complaint patterns, examine the factors associated with slower resolution and financial recovery, and generate evidence-based insights that can support better insurance operations and customer service.

- Understand the dataset structure and complaint attributes.
- Clean and prepare the data, handling missing values and duplicates.
- Perform Exploratory Data Analysis (EDA) and statistical analysis.
- Engineer new features such as Resolution Days and Recovery Status.
- Identify complaint patterns and resolution trends through visualizations.
- Generate actionable business insights.

3. Problem Statement

Insurance companies receive thousands of complaints every year but often struggle to identify patterns in complaint types, resolution speed and recovery rates. Without proper analysis, it is difficult to understand which complaint categories take longer to resolve and what factors influence recovery outcomes.

This project investigates which companies, coverage types and complaint reasons account for the most complaints, how quickly complaints are resolved, and how often and how much money is recovered. The dataset records what happened to each complaint but not the underlying cause of each dispute, so the analysis identifies associations and areas for further investigation rather than proving causes.

4. Dataset Information

- **Dataset:** Insurance Company Complaints, Resolutions, Status, and Recoveries
- **Source:** U.S. Government Open Data Portal (data.gov)
- **Domain:** Insurance and Consumer Complaint Analytics
- **File format:** CSV
- **Dataset size:** 500 complaint records (sampled from 89,348 original records) and 12 original columns
- **Complaint period:** 2021–2026 (one record in the sample was opened in 2018); the latest record was opened on 5 October 2026
- **Analysis environment:** Google Colab; **Programming language:** Python
- **Outcome variables:** Status (open/closed), Resolution Days (derived) and Recovery (USD)

5. Dataset Attributes

Attribute	Data Type	Description
Company	Object	Name of the insurance company
File No.	Integer	Unique complaint file number
Opened	Date	Date the complaint was opened
Closed	Date	Date the complaint was closed
Coverage	Object	Type of insurance coverage
SubCoverage	Object	Sub-type of insurance coverage
Reason	Object	Primary reason for the complaint
SubReason	Object	Detailed reason for the complaint
Disposition	Object	How the complaint was handled
Conclusion	Object	Final conclusion of the complaint
Recovery	Float	Amount recovered, in USD
Status	Object	Current complaint status

6. Tools and Technologies

- Python – programming and analysis
- Pandas – data loading, cleaning, grouping and summaries
- NumPy – numerical operations
- Matplotlib – chart creation
- Seaborn – statistical visualizations
- Google Colab – notebook development and execution
- GitHub – source-code and dataset storage

7. Project Workflow

Stage 1: Initial Exploratory Data Analysis

The dataset was loaded from GitHub into a Pandas DataFrame (shape 500 × 12). Initial exploration included previewing records with `df.head()`, checking column names and data types with `df.info()`, generating descriptive statistics, counting unique values per column, and checking missing values (513 across 7 columns) and duplicate records (0 found).

Stage 2: Data Cleaning, Transformation and Feature Engineering

Date columns (Opened, Closed) were converted to datetime format. In the Recovery column the `$` and `,` symbols were removed and the values converted to numeric. The column `File No.` was renamed `File_No`. No duplicates were found, so no rows were dropped. Missing values were treated as follows, leaving zero missing values:

Column	Missing	Share	Method Used
Disposition	207	41.4%	Filled with “Unknown”
Conclusion	137	27.4%	Filled with “Unknown”
SubCoverage	90	18.0%	Filled with “Not Specified”
Coverage	25	5.0%	Filled with mode
Reason	20	4.0%	Filled with mode
SubReason	20	4.0%	Filled with mode
Closed	14	2.8%	Filled with the Opened date

Eight new features were then engineered, taking the dataset from 12 to **20 columns** (500 rows × 20 columns):

New Feature	Definition
Resolution_Days	Days between Opened and Closed dates
Resolution_Category	Fast (≤ 15 days), Medium (16–30 days), Slow (> 30 days)
Recovery_Status	“Recovered” if Recovery > 0, otherwise “Not Recovered”
Is_Recovered	Binary flag (1 = Recovered, 0 = Not Recovered)
Year, Month, Quarter	Extracted from the Opened date for time-based analysis
Complaint_Age	Old (2021–22), Recent (2023–24), Latest (2025–26)

Stage 3: Statistical Analysis and Visualization

Statistical analysis included measures of central tendency, variance, standard deviation, skewness, kurtosis and outlier detection using the IQR method. Ten objective-based visualizations were created: four univariate, three bivariate and three multivariate. Each visualization in Section 9 includes the objective, visualization, interpretation, key insight and business insight.

Note: the charts in Section 9 are the original charts produced in the project’s Google Colab notebook.

8. Statistical Analysis

8.1 Measures of Central Tendency

Objective: To calculate the mean, median and mode of Resolution Days and Recovery to understand typical complaint behavior.

Variable	Mean	Median	Mode	Min	Max
Resolution Days	42.06	28.50	21	0	880
Recovery (USD), all complaints	1,756.72	0.00	0.00	0.00	338,964.51
Recovery (USD), recovered only (n = 97)	9,055.25	1,301.08	—	9.12	338,964.51

Interpretation: The mean resolution time (42.06 days) is well above the median (28.50 days), meaning a small number of very long cases pull the average upward. The median and mode of Recovery are both zero because most complaints (80.6%) recovered no money; among the 97 complaints that did recover money, the median recovery was 1,301.08 USD and the mean 9,055.25 USD.

Key Insight: The typical complaint is resolved in about four weeks and recovers nothing; the average is shaped by a few long cases and a few large recoveries.

Business Insight: The median is a more reliable measure of typical service performance than the mean for both variables. Service-level targets should be set on median and percentile values, with long-running cases tracked separately.

8.2 Variance and Standard Deviation

Objective: To measure the spread of Resolution Days and Recovery around their respective means.

Variable	Variance	Standard Deviation
Resolution Days	3,117.11	55.83
Recovery (USD)	278,912,716.50	16,700.68

Interpretation: Resolution times typically vary by about 56 days around the 42-day mean, larger than the mean itself, which signals a strongly right-skewed distribution. The standard deviation of Recovery (16,700.68 USD) is almost ten times its mean (1,756.72 USD), reflecting highly uneven recovery amounts.

Key Insight: Both measures are highly dispersed. Standard deviations should be read in each variable’s own units (days versus USD) and not compared directly.

Business Insight: Averages alone can mislead for complaint handling. Reporting percentiles (for example, the share of complaints closed within 30 days) gives management a clearer picture.

8.3 Skewness, Kurtosis and Outlier Detection

Objective: To examine distribution shape and identify potential outliers in Resolution Days and Recovery.

Variable	Skewness	Excess Kurtosis
Resolution Days	7.898	103.773
Recovery (USD)	17.443	339.055

IQR analysis of Resolution Days	Value
Q1 (25th percentile)	18.00 days
Q3 (75th percentile)	47.25 days
IQR	29.25 days
Upper bound (Q3 + 1.5 × IQR)	91.13 days
Potential outliers above the upper bound	43 complaints (8.6%)

Interpretation: Both variables are extremely right-skewed with very heavy tails. For Resolution Days, 43 complaints took longer than 91.13 days, and the longest case took 880 days; this record was opened in March 2018 and closed in August 2020, well outside the 2021–2026 period. The largest single recovery (338,964.51 USD) alone accounts for 38.6% of the total recovery of 878,358.86 USD.

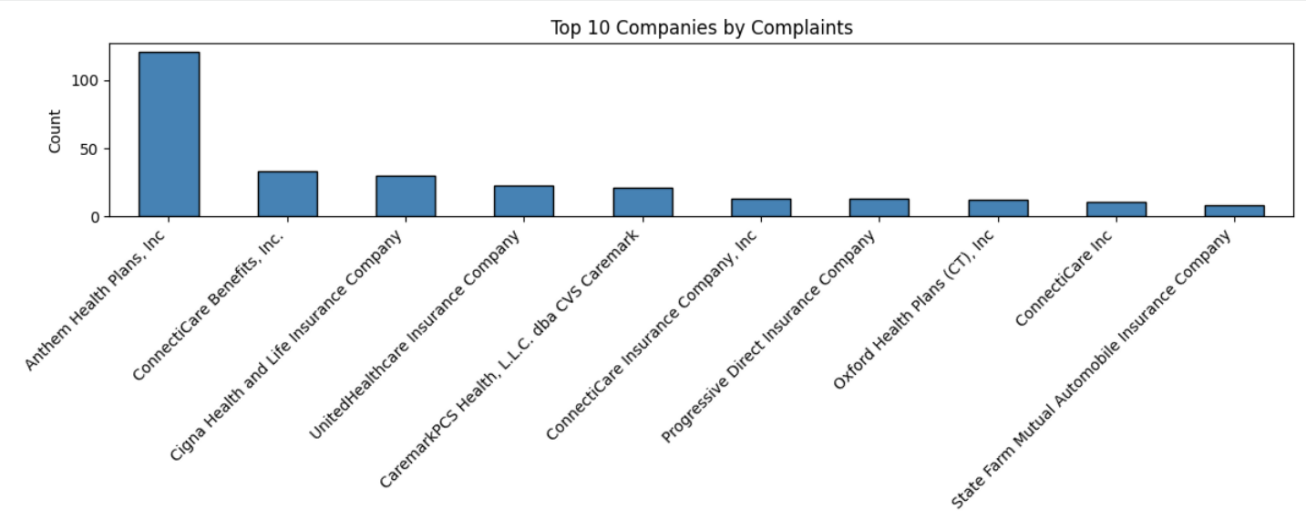
Key Insight: A handful of extreme records drive the averages of both variables. The long-running cases are the main source of delay; the large recovery is a one-off that dominates the total.

Business Insight: Retain these observations unless they are confirmed data errors, since they may represent genuine long-running disputes. Review the aging cases individually and treat large recoveries separately when estimating typical outcomes.

9. Objective-Based Visualizations and Interpretations

Univariate Analysis

Visual 1 — Top 10 Companies by Complaint Count



Objective: To identify which insurance companies received the highest number of complaints.

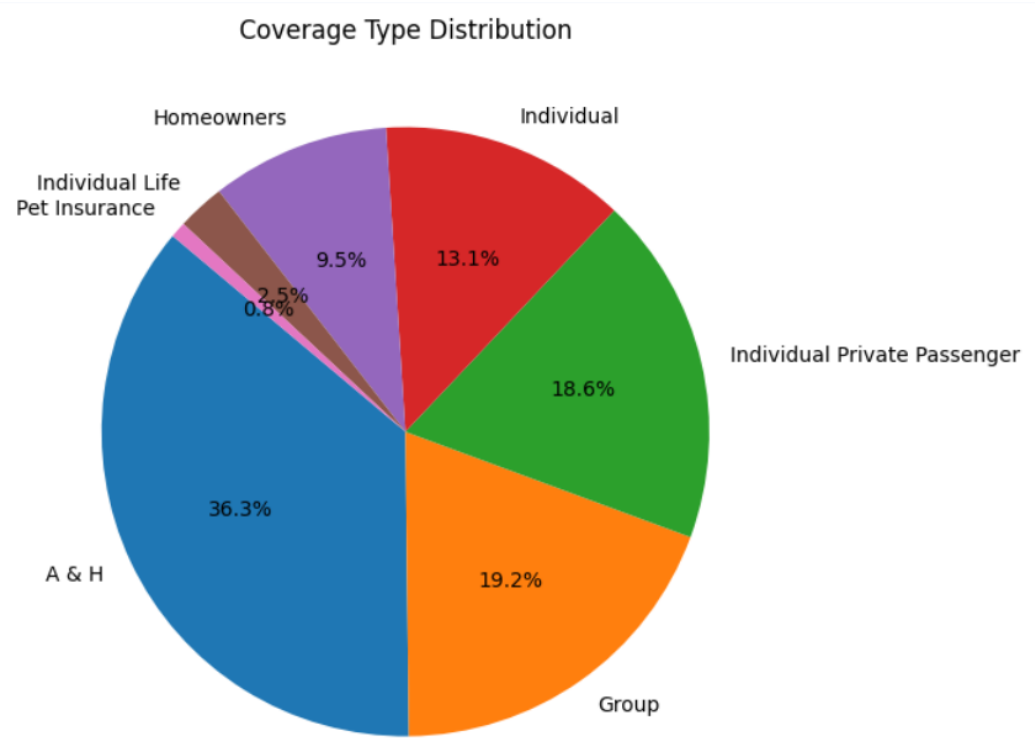
Visualization: Visual 1 — Top 10 Companies by Complaint Count (bar chart from the Python notebook).

Interpretation: Anthem Health Plans, Inc leads with 121 complaints (24.2% of the 500 records), followed by ConnectiCare Benefits, Inc. (33), Cigna Health and Life Insurance Company (30) and UnitedHealthcare Insurance Company (23). The dataset contains 136 distinct company names.

Key Insight: One company alone accounts for nearly one-quarter of all complaints in the dataset, more than three times the next company.

Business Insight: Anthem Health Plans should be prioritized for a root-cause review of its complaints. Raw counts are not adjusted for company size or number of policies, so complaint *rates* should be compared before drawing conclusions about service quality.

Visual 2 — Coverage Type Distribution



Objective: To understand which insurance coverage types receive the most complaints.

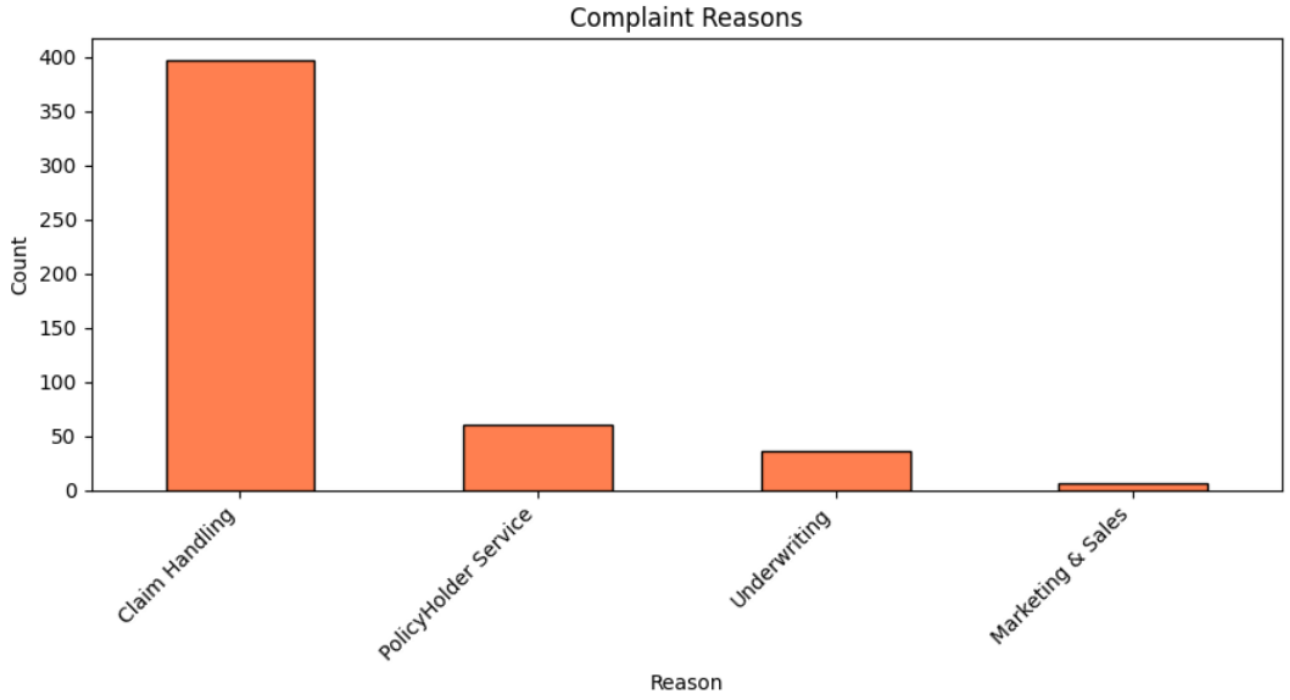
Visualization: Visual 2 — Coverage Type Distribution (pie chart from the Python notebook).

Interpretation: The pie chart shows each coverage type as a share of the seven largest types in the cleaned data: A & H 36.3%, Group 19.2%, Individual Private Passenger 18.6%, Individual 13.1%, Homeowners 9.5%, Individual Life 2.5% and Pet Insurance 0.8%. Measured against all 500 complaints, A & H accounts for 29.4% (147 complaints) as originally recorded; 25 records had no coverage and were filled with the most common type, which raises A & H to 172 complaints (34.4%) in the cleaned data.

Key Insight: A & H coverage has the highest complaint share on any basis, and A & H, Group and Individual Private Passenger together make up about three-quarters of the complaints in the chart.

Business Insight: Insurers offering A & H coverage should review their claim-handling procedures to reduce complaint volumes.

Visual 3 — Complaint Reasons Distribution



Objective: To identify the most frequently occurring complaint reasons.

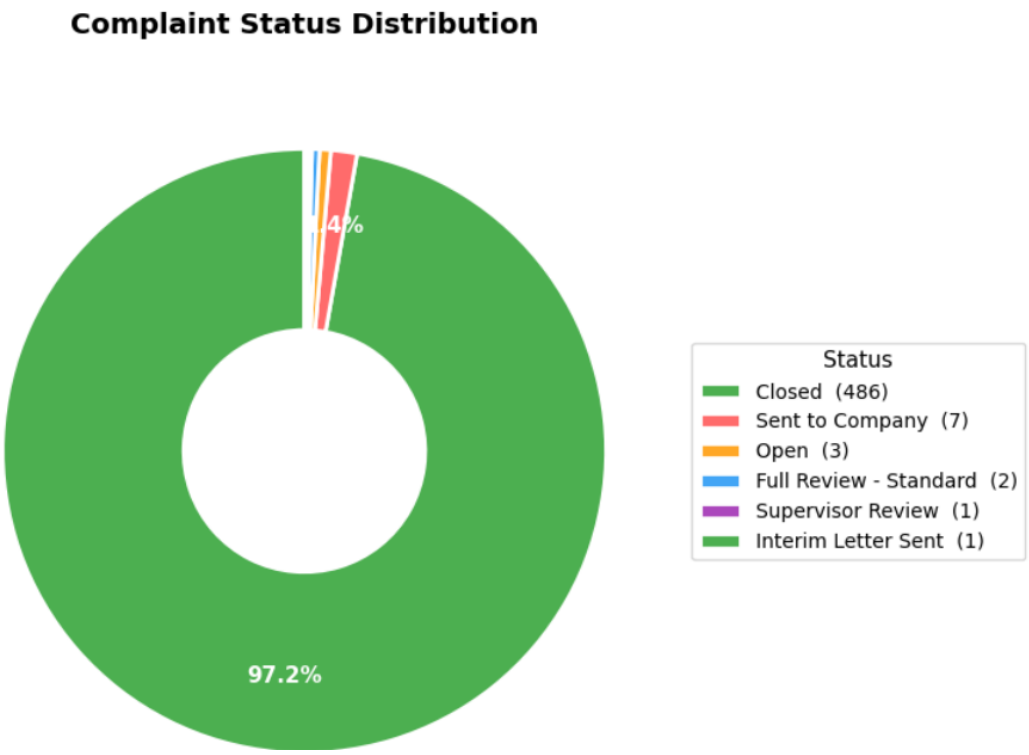
Visualization: Visual 3 — Complaint Reasons Distribution (bar chart from the Python notebook).

Interpretation: Claim Handling is the dominant reason, followed by PolicyHolder Service (61), Underwriting (36) and Marketing & Sales (6). As originally recorded, Claim Handling accounts for 377 of 500 complaints (75.4%); the reason was missing for 20 complaints (4.0%), which were filled with the most common reason. The chart uses the cleaned data, so its Claim Handling bar shows about 397 complaints (79.4%).

Key Insight: Roughly three out of four complaints (75.4% as recorded) relate to how claims are handled.

Business Insight: Insurers should prioritize improving claim-processing speed, transparency and communication with policyholders.

Visual 4 — Complaint Status Distribution



Objective: To monitor how many complaints are closed versus still in progress.

Visualization: Visual 4 — Complaint Status Distribution (donut chart from the Python notebook).

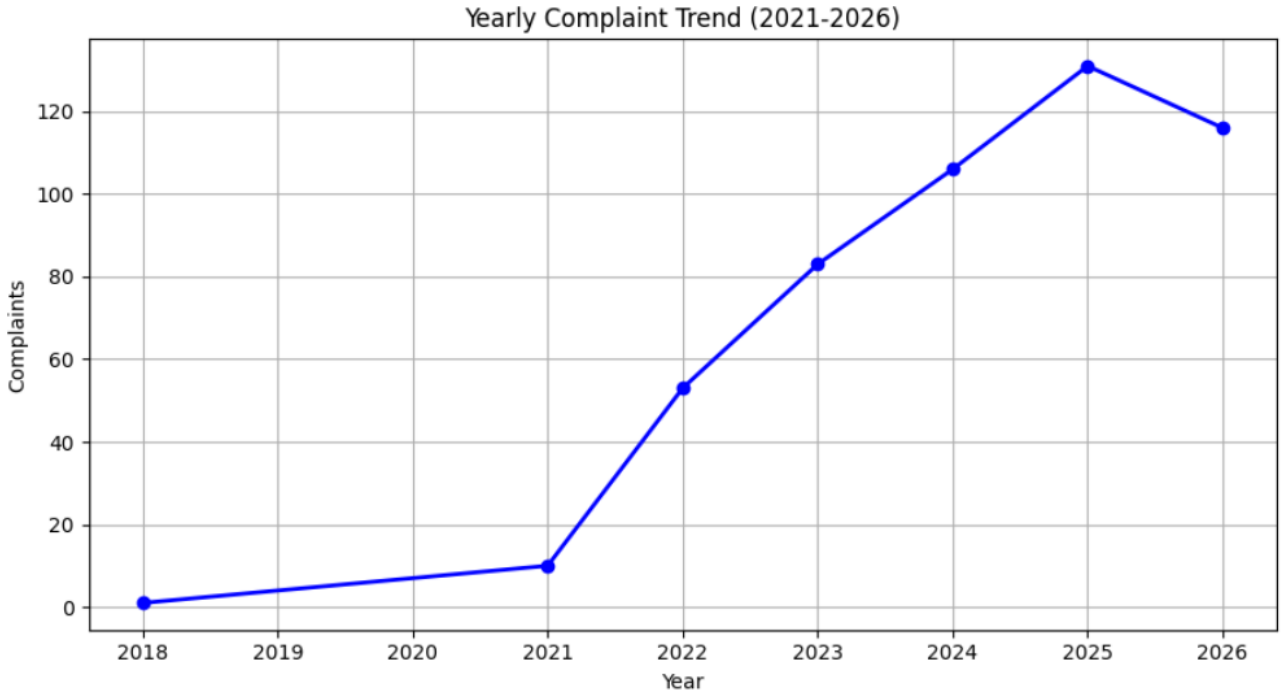
Interpretation: Of 500 complaints, 486 (97.2%) are closed. The remaining 14 (2.8%) are Sent to Company (7), Open (3), Full Review – Standard (2), Supervisor Review (1) and Interim Letter Sent (1).

Key Insight: Nearly all complaints in the dataset have reached closure.

Business Insight: The 14 complaints not yet closed should be monitored and escalated so that they do not age further.

Bivariate Analysis

Visual 5 — Yearly Complaint Trend (2021–2026)



Objective: To analyze how the number of complaints has changed from 2021 to 2026.

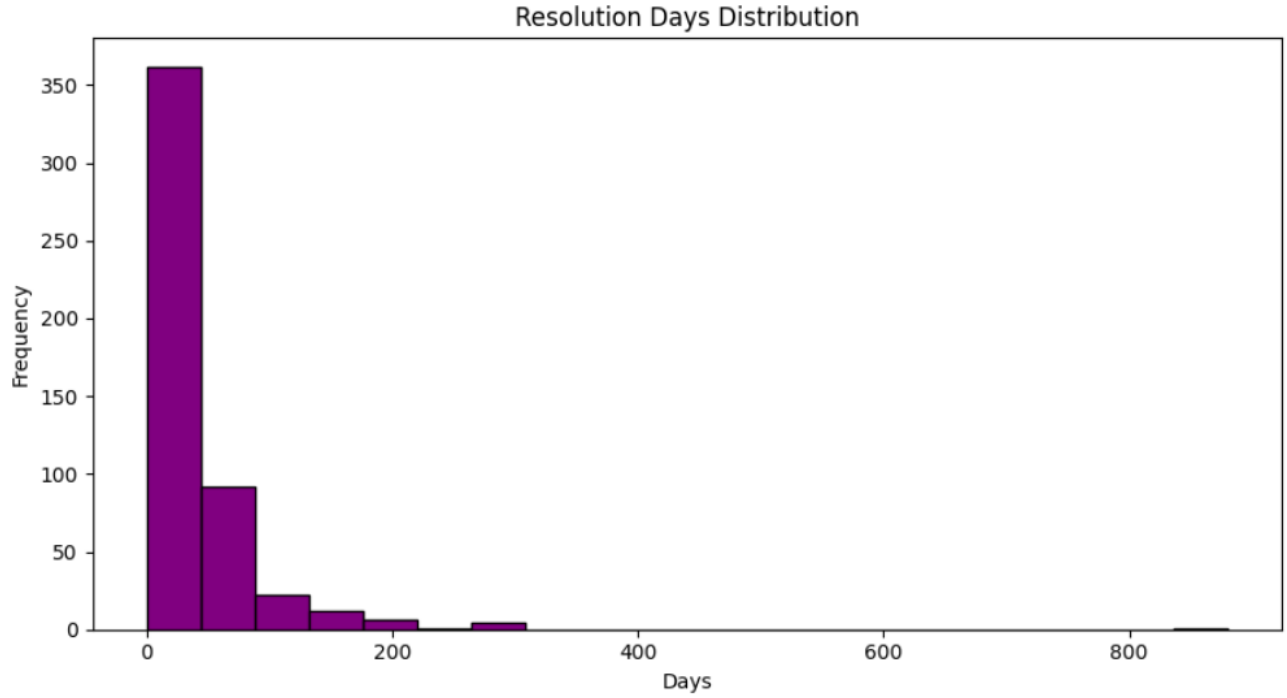
Visualization: Visual 5 — Yearly Complaint Trend (2021–2026) (line chart from the Python notebook).

Interpretation: Complaints rose every year from 10 in 2021 to 53 (2022), 83 (2023), 106 (2024) and a peak of 131 in 2025 — an increase of more than 1,200% from 2021 to 2025. The 116 complaints for 2026 cover the period only up to 5 October 2026, so the year is incomplete. The chart also shows a single complaint opened in 2018 (an isolated record outside the 2021–2026 period), which is why the axis starts at 2018.

Key Insight: Complaint volume in the sample has grown steadily each full year since 2021.

Business Insight: Because the data is a 500-record sample of 89,348 records, the trend should be confirmed on the full dataset before it is read as a true rise in complaints. If confirmed, proactive intervention in complaint-heavy periods could help retain customers.

Visual 6 — Resolution Days Distribution



Objective: To understand the distribution of the time taken to resolve complaints.

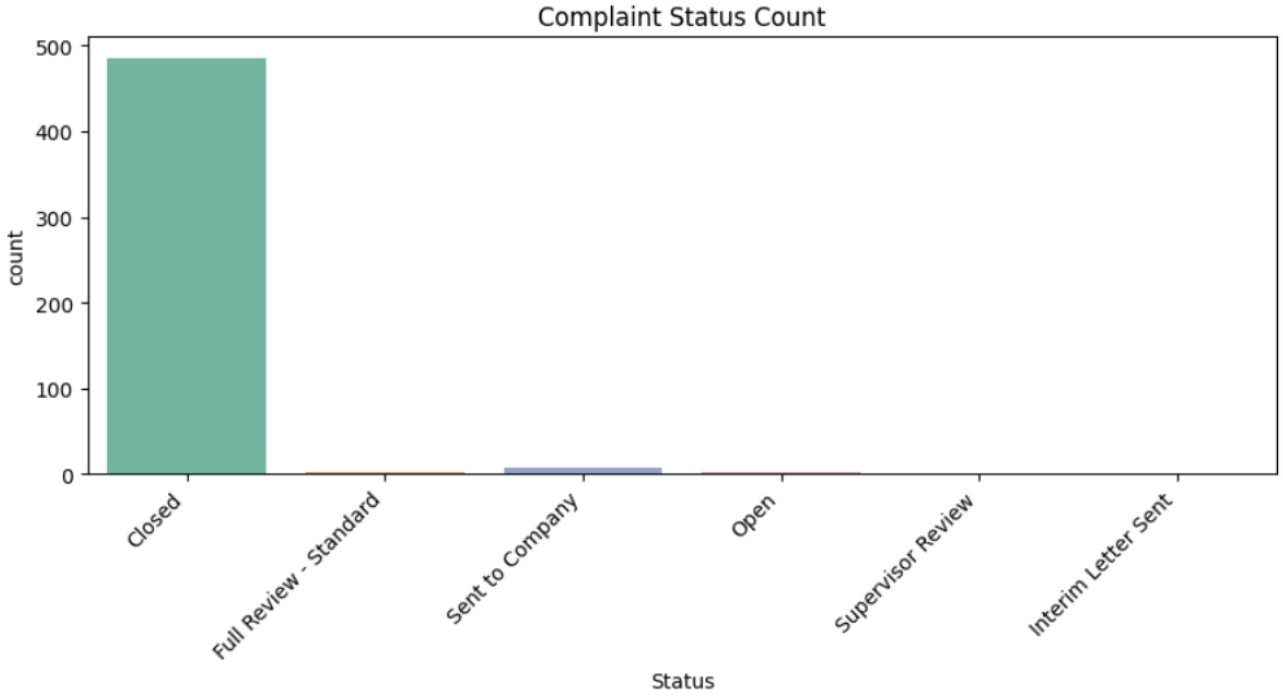
Visualization: Visual 6 — Resolution Days Distribution (histogram from the Python notebook).

Interpretation: The histogram uses bins of about 44 days. The first bin (0–44 days) holds 362 complaints (72.4%) and the second (44–88 days) holds 92, after which counts fall away, with a long thin tail reaching one case at 880 days. The median is 28.5 days and the mean 42.06 days. By category, 103 complaints (20.6%) were Fast (≤ 15 days), 160 (32.0%) Medium (16–30 days) and 237 (47.4%) Slow (> 30 days).

Key Insight: Nearly half (47.4%) of complaints take longer than 30 days to resolve.

Business Insight: Companies should aim to bring typical resolution time below 30 days by streamlining internal review processes, and should review the small number of very long cases.

Visual 7 — Complaint Status Count



Objective: To compare the exact counts of each complaint status.

Visualization: Visual 7 — Complaint Status Count (count plot from the Python notebook).

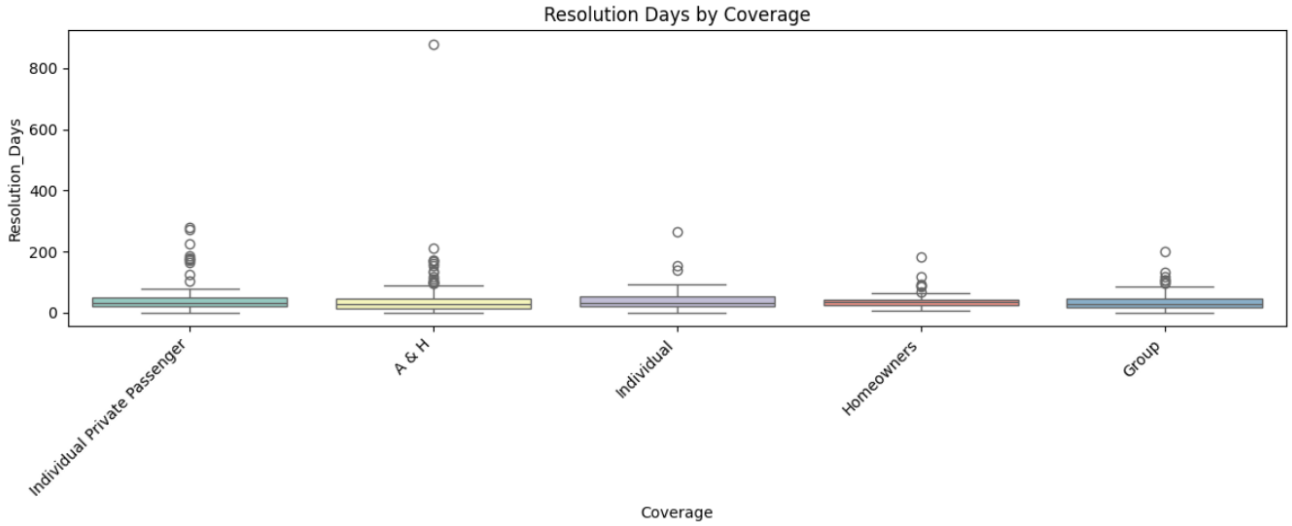
Interpretation: 486 complaints are Closed. The remaining 14 are: Sent to Company (7), Open (3), Full Review – Standard (2), Supervisor Review (1) and Interim Letter Sent (1).

Key Insight: The dataset reflects strong closure performance, with only 2.8% of complaints still in progress.

Business Insight: Focus should shift from volume closure to the quality of resolution and customer satisfaction.

Multivariate Analysis

Visual 8 — Resolution Days by Coverage Type



Objective: To compare resolution speed across the five insurance coverage types with the most complaints.

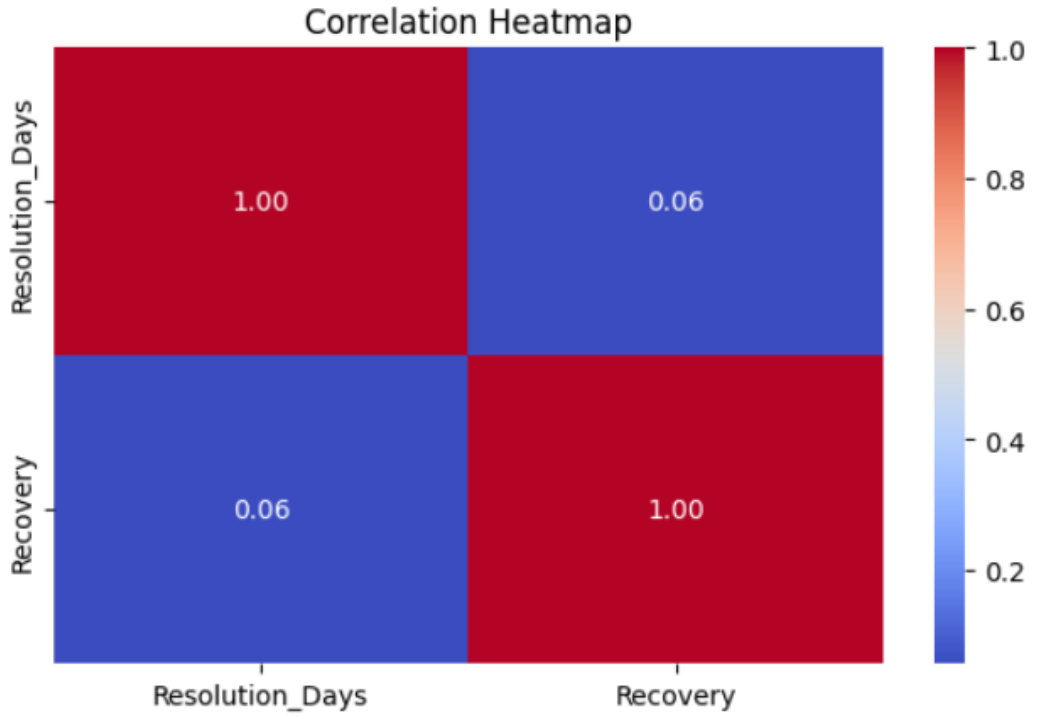
Visualization: Visual 8 — Resolution Days by Coverage Type (box plot from the Python notebook).

Interpretation: The box plot compares the five coverage types with the most complaints. Individual Private Passenger has the slowest average resolution at 49.43 days, followed by A & H (42.83), Homeowners (41.49), Individual (41.21) and Group (36.01). Medians are much lower (27–34 days) and the boxes overlap widely, with outliers up to about 280 days in Individual Private Passenger. The single point near 880 days under A & H is the 2018 complaint. Individual Life averages 61.5 days but has only 12 complaints and is not shown.

Key Insight: Individual Private Passenger is the slowest of the larger coverage types, but the gap between coverage types is modest compared with the variation within each type.

Business Insight: Coverage types with consistently high resolution times, particularly Individual Private Passenger, could be assigned dedicated fast-track teams.

Visual 9 — Correlation Matrix — Resolution Days and Recovery



Objective: To identify the statistical relationship between resolution time and financial recovery.

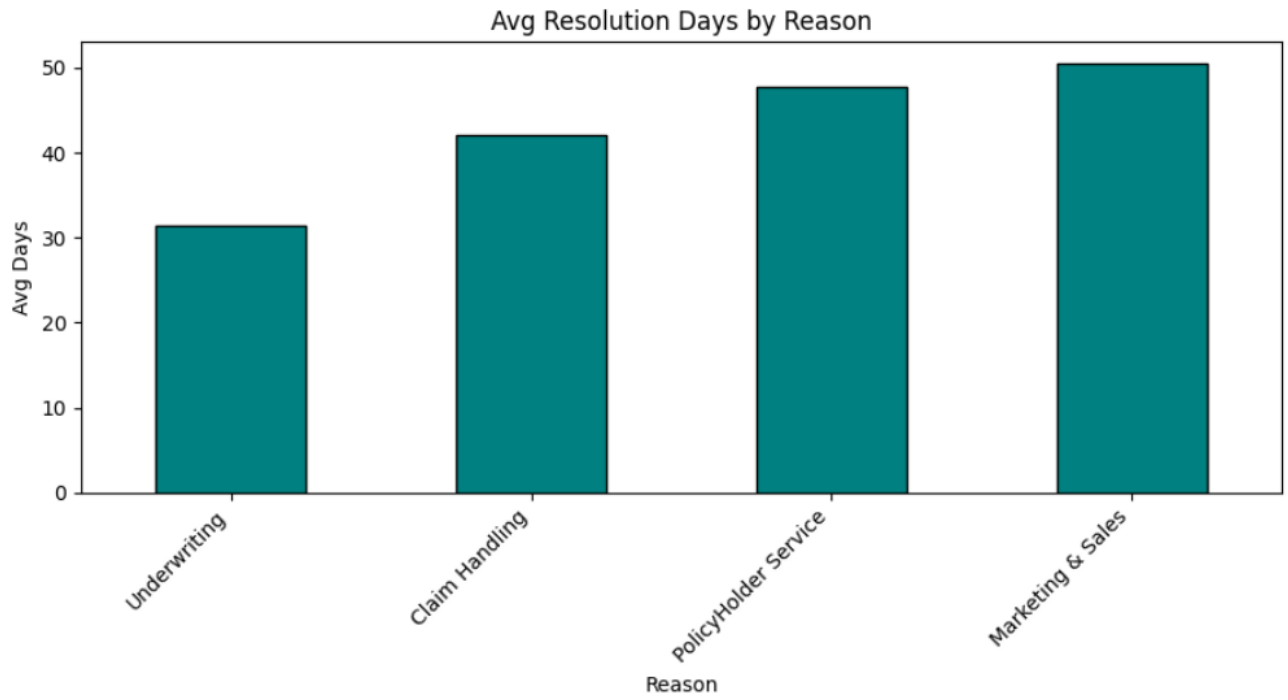
Visualization: Visual 9 — Correlation Matrix — Resolution Days and Recovery (heatmap from the Python notebook).

Interpretation: The heatmap shows a correlation of only 0.06 (0.057 before rounding) between Resolution_Days and Recovery. Values near zero mean the two variables move almost independently of each other.

Key Insight: Resolution speed does not meaningfully predict how much money is recovered.

Business Insight: Recovery outcomes appear to depend more on the complaint type and coverage than on speed; these categories are better starting points for estimating recovery probability.

Visual 10 — Average Resolution Days by Complaint Reason



Objective: To identify which complaint reasons take the longest on average to resolve.

Visualization: Visual 10 — Average Resolution Days by Complaint Reason (bar chart from the Python notebook).

Interpretation: Marketing & Sales complaints take the longest at an average of 50.50 days, followed by PolicyHolder Service (47.75), Claim Handling (42.02) and Underwriting (31.39). Marketing & Sales has only 6 complaints, so its average is less stable than the others.

Key Insight: Different complaint reasons have noticeably different average resolution timelines; PolicyHolder Service is the slowest reason with a substantial volume (61 complaints).

Business Insight: Insurers should create reason-specific service-level targets and escalation paths for categories with above-average resolution times.

10. Overall Key Findings

Metric	Result
Most common complaint reason	Claim Handling — 75.4% of complaints
Top coverage type	Accident & Health (A & H) — 29.4%
Top complaint company	Anthem Health Plans, Inc — 121 complaints (24.2%)
Closed complaints	486 of 500 (97.2%)
Average resolution time	42.06 days (median 28.5); 47.4% take longer than 30 days
Complaint volume trend	10 in 2021 to 131 in 2025
Slowest reason (average)	Marketing & Sales — 50.50 days (only 6 complaints)
Total financial recovery	878,358.86 USD across 97 complaints (19.4%)
Recovery concentration	Claim Handling complaints account for 85.7% of all recovered money
Resolution speed vs. recovery	Minimal correlation (r = 0.057)

11. Business Recommendations

1. Conduct a root-cause review of Anthem Health Plans' complaints, comparing its complaint *rate* against other companies before setting targets.
2. Prioritize improvements in claim handling — processing speed, transparency and communication — since it drives 75.4% of complaints and most recovered money.
3. Set reason-specific and coverage-specific service levels, with fast-track handling for Individual Private Passenger and PolicyHolder Service complaints, and aim to bring typical resolution below 30 days.
4. Review long-running cases individually (the 43 complaints above 91 days) and monitor the 14 complaints that are not yet closed to prevent further aging.
5. Track complaint volume, resolution time and recovery rate over time, and measure the effect of any process change on these indicators.

12. Limitations

- The analysis uses a 500-record sample of 89,348 original records; the sampling method is not stated, so results may not represent the full dataset.
- The dataset records outcomes, not the underlying cause of each dispute, so the analysis identifies associations rather than causal relationships.
- Complaint counts are not adjusted for company size or policy volume, so high counts do not by themselves indicate poor service.
- Missing values were filled using simple methods: 41.4% of Disposition and 27.4% of Conclusion values are "Unknown", and mode imputation slightly increases the dominant categories of Coverage, Reason and SubReason. The 14 missing Closed dates were set to the Opened date, which gives those complaints a resolution time of zero days. Percentages quoted in the text for Coverage and Reason are as originally recorded, while the charts use the cleaned data.
- The year 2026 is incomplete (data runs to 5 October 2026), and one record opened in 2018 is an extreme outlier (880 days).

- Some groups are very small (for example, 6 Marketing & Sales complaints), so their averages are less reliable.
- No predictive machine-learning model was built or evaluated in this project.

13. Future Enhancements

1. Include more records from all U.S. states for better geographic accuracy.
2. Normalize complaint counts by company size to compare complaint rates fairly.
3. Conduct statistical significance tests for selected group comparisons.
4. Apply machine-learning models to predict complaint resolution time and recovery outcomes.
5. Build interactive dashboards using Power BI or Tableau.
6. Apply NLP to complaint reason text for deeper insights.
7. Automate reporting with scheduled Python scripts.

14. Conclusion

The Insurance Company Complaints project demonstrates how Python, statistical analysis and visualization can be used to understand complaint patterns, resolution trends and recovery behavior. The sample of 500 complaints shows that Claim Handling is the dominant complaint reason (75.4%), that Accident & Health is the most-complained-about coverage (29.4%), and that one company, Anthem Health Plans, accounts for 24.2% of complaints. Complaint volumes in the sample grew from 10 in 2021 to 131 in 2025.

Almost all complaints (97.2%) are closed, but resolution is uneven: the median complaint takes 28.5 days while the average is 42.06 days, and 47.4% take longer than 30 days because of a small number of very long cases. Financial recovery was achieved in 19.4% of complaints, totaling 878,358.86 USD; it is concentrated in a few large recoveries and is only weakly related to resolution speed.

Because the dataset is a sample and records outcomes rather than causes, the findings should be read as associations and as pointers for further investigation. Overall, the project demonstrates practical skills in data inspection, cleaning, feature engineering, exploratory and statistical analysis, outlier detection, visualization and business-focused reporting. Future work could extend it to the full dataset, normalized complaint rates and predictive models.

Appendix: Repository and How to Run

File / Folder	Purpose
Insurance Company.csv	Sampled dataset (500 rows)
Insurance_Complaints.ipynb	Google Colab notebook
screenshots/	Stage and visualization screenshots
README.md	Project documentation

To run: open Google Colab, upload `Insurance_Complaints.ipynb`, and choose Runtime → Run all. All outputs, charts and insights are generated automatically.

Data source and license: U.S. Government Open Data Portal (data.gov); used for educational and analytical purposes only. Repository: github.com/Kumarsakthi835/Kumarsakthi835-Insurance-Company-Complaints-Resolutions-Status-and-Recoveries